

GTU CSE 241 / 501 Quiz

Std Code: 158

Std Number:

2	1	0	1	0	4	0	0	4	0	1	7			
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```
double netArea(const vector<Shape* >& v ) {  
    double tot_sqr = 0.0;  
    double other = 0.0;  
    if( v.empty ( ) ) exit(1) ;  
    for(int i = 0 ; i< v.size() ; i++ ) { // normal loop  
        Square *temp = dynamic_cast<Shape *> ( v[i] );  
        if( temp !=nullptr) tot_sqr += v[i]->area();  
        else other += v[i]->area();  
  
    }  
  
    /* if there are getters , other solution is  
        for( const auto& temp : v ) {  
            if(temp->get_len() == temp-> get_width() )  
                tot_sqr += temp->area();  
            else other +=temp->area();  
        }    */  
  
    return tot_sqr – other;  
  
}
```

```
int main( ) {  
    vector<Shape* >vec( 5 ); // at least five  
    Circle cir(3.0);  
    Square sq (3.0 , 3.0);  
    Rectangle rec(2.0 , 3.0);  
    Circle cir_2(5.0);  
    Square sq_2(4.0 , 4.0 );  
    vec[0] =&cir;  
    vec[1] =&sq;  
    vec[2] =&rec;  
    vec[3] =&cir_2;  
    vec[4] =&sq_2;  
    //I could pushback also , but I chose that way.  
    double result = netArea(vec);  
    cout << “ the result is -> “ << result << endl;  
    return 0;  
}
```

```
int main( ) {
```

```
    return 0;
```

```
}
```